Emmanuel Sifiso Nkosi

Data Scientist

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Professional Summary

Data Scientist with 3 years of experience building, training, and deploying machine learning and deep learning models. Analysing large-scale datasets, leveraging AI and LLM APIs, and delivering production-ready solutions across cloud and full-stack environments.

Experience

Wits Enterprise (University of the Witwatersrand) 12/2024 – Present | Johannesburg

Data Scientist

- Developing AI solutions leveraging LLMs, NLP, and Deep Learning to uncover patterns and analyze large-scale social media datasets and inform business decisions in telecommunications.
- Applying advanced unsupervised learning techniques such as UMAP for dimensionality reduction and HDBSCAN for clustering, combined with BERTopic, LLAMA2, and OpenAI models to extract, label, and interpret latent topics in unstructured datasets.
- Integrated ElevenLabs AI voice with a report-based chatbot to enable natural, conversational querying of qualitative analyst reports.
- Applied Quantum Machine Learning (QML) techniques on financial data using SamplerQNN and EstimatorQNN, exploring hybrid quantum-classical modeling approaches.
- Expanding into frontend engineering using Angular and React to productize internal tools.
- Integrating API endpoints from Swagger exposed by backend team UI components to SQL Server via token-authenticated endpoints.
- Continuously deploying to a live application hosted on Azure Web Apps

Telkom Corporate Centre 01/2024 – 12/2024 | Pretoria (Centurion)

Junior Data Scientist

- Worked on IoT-integrated Digital Twin proof-of-concepts for predictive maintenance and system monitoring.
- Simulated telemetry data using AWS IoT SiteWise and 3D motor models through TwinMaker.
- Trained and deployed ML models on AWS SageMaker and EC2 to analyze and forecast equipment conditions.
- Built real-time dashboards with Grafana using data stored in S3.
- Led interns through end-to-end acoustic-modeling and machine-learning projects, driving technical direction, model development, and applied ML practices.

BCX (Business Connexion) 01/2023 – 12/2023 | Pretoria (Centurion)

Data Scientist Intern

- Built a Speech Sentiment Analysis system to monitor customer satisfaction and agent performance in call centers.
- Performed data labeling, EDA, preprocessing (MFCCs, spectrograms), and augmentation.
- Trained deep learning models including CNN, RNN, and Transformers, deployed and presented the project.

Education

IT Varsity 2025 – 2025

Full-stack Development NQF 5 Certificate

- GitHub | Principles of UX design | Design Thinking | API's + Data Input + Processing | User Centric App Development | Data Management and Analysis | Backend Development (Building an API)

Explore AI 2023 – 2023

Nationa Certificate: Information Technology-Systems Development NQF5 (Specialization: Data Science)

- Power BI | Python | Data Structures & Algorithms | SQL | Machine Learning

Vukuzame MST School 2019 – 2019

Matric Certificate (Science)

Skills

Languages & Tools
Python, SQL, Angular, Node.js, React.js

Libraries & Frameworks
Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenAI, LangChain, React, Angular

ML & AI
Predictive Modeling, LLMS, RAG Pipelines, NLP, Deep Learning, CNNS, RNNS, Transformers, Flowise

Deployment & DevOps
AWS EC2, Azure Web Apps, Render (CI/CD), Vercel, GitHub Pages, GitHub Actions

APIs & Databases
RESTful APIs, Swagger, SSMS (SQL Server), PostgreSQL, Vector Database (Pinecone)

Additional

Driver's Licence

- Code: 10 (C1)
- Status: Valid & Upto Date